Project Report:

Employee Attrition Prediction

Predictive Analytics Project

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Executive Summary

This project analyzes employee attrition using the IBM HR Analytics Employee Attrition dataset and compares multiple machine learning models in R to identify the strongest predictors of employee turnover. The analysis includes data exploration, correlation review, attrition trend visualizations and model development using Random Forest, Gradient Boosting, CART and Logistic Regression.

The results show that overtime, job role, monthly income, total working years and years at company are among the most important factors associated with attrition. Logistic Regression and Gradient Boosting performed best overall based on the project's evaluation metrics, while Random Forest also showed strong predictive capability.

The findings suggest that organizations can reduce attrition by focusing on overtime management, early-career employee support, targeted retention strategies for high-turnover roles and better support for employees facing longer commutes. This project demonstrates how predictive analytics can support proactive HR decision-making and improve workforce retention strategy.

1. Introduction

Employee turnover is a significant challenge for many organizations, impacting operational efficiency and incurring high costs in recruitment and training. High attrition rates can disrupt productivity, lead to loss of organizational knowledge, and increase hiring expenses. This project aims to develop a predictive analytics model to identify factors contributing to employee attrition and implement strategies to retain valuable employees. Early identification and intervention can improve employee satisfaction, reduce turnover rates, and enhance overall organizational performance.

2. Dataset

Dataset Source: https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset

This project utilizes a dataset from Kaggle containing anonymized HR data. The dataset includes:

- **Employee Demographics:** Age, gender, marital status, etc.
- **Performance Metrics:** Performance rating, years at the company, etc.
- **Compensation Details:** Monthly income, stock option level, etc.
- **Job Satisfaction Surveys:** Job satisfaction, work-life balance, environment satisfaction, etc.
- **Historical Turnover Records:** Attrition status, number of companies worked, etc.

Data Preprocessing Steps:

- Initial exploration includes checking dimensions, missing values, and duplicate rows.
- Checked for and handled any missing values and duplicate rows.

```
> # Explore the dataset
> dim(df)
[1] 1470   35
> sum(is.na(df))
[1] 0
> any(duplicated(df))
[1] FALSE
```

- Basic statistical summaries and descriptions are generated.

```
> # Basic statistics and summary
> describe(df)
```

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis
Age	1	1470	36.92	9.14	36.0	36.47	8.90	18	60	42	0.41	-0.41
Attrition*	2	1470	1.16	0.37	1.0	1.08	0.00	1	2	1	1.84	1.39
BusinessTravel*	3	1470	2.61	0.67	3.0	2.76	0.00	1	3	2	-1.44	0.69
DailyRate	4	1470	802.49	403.51	802.0	803.83	510.01	102	1499	1397	0.00	-1.21
Department*	5	1470	2.26	0.53	2.0	2.25	0.00	1	3	2	0.17	-0.40
DistanceFromHome	6	1470	9.19	8.11	7.0	8.08	7.41	1	29	28	0.96	-0.23
Education	7	1470	2.91	1.02	3.0	2.98	1.48	1	5	4	-0.29	-0.56
EducationField*	8	1470	3.25	1.33	3.0	3.10	1.48	1	6	5	0.55	-0.69
EmployeeCount	9	1470	1.00	0.00	1.0	1.00	0.00	1	1	0	NaN	NaN
EmployeeNumber	10	1470	1024.87	602.02	1020.5	1023.40	790.97	1	2068	2067	0.02	-1.23
EnvironmentSatisfaction	11	1470	2.72	1.09	3.0	2.78	1.48	1	4	3	-0.32	-1.20
Gender*	12	1470	1.60	0.49	2.0	1.62	0.00	1	2	1	-0.41	-1.83
HourlyRate	13	1470	65.89	20.33	66.0	66.02	26.69	30	100	70	-0.03	-1.20
JobInvolvement	14	1470	2.73	0.71	3.0	2.74	0.00	1	4	3	-0.50	0.26
JobLevel	15	1470	2.06	1.11	2.0	1.90	1.48	1	5	4	1.02	0.39
JobRole*	16	1470	5.46	2.46	6.0	5.61	2.97	1	9	8	-0.36	-1.20
JobSatisfaction	17	1470	2.73	1.10	3.0	2.79	1.48	1	4	3	-0.33	-1.22
MaritalStatus*	18	1470	2.10	0.73	2.0	2.12	1.48	1	3	2	-0.15	-1.12
MonthlyIncome	19	1470	6502.93	4707.96	4919.0	5667.24	3260.24	1009	19999	18990	1.37	0.99
MonthlyRate	20	1470	14313.10	7117.79	14235.5	14286.48	9201.76	2094	26999	24905	0.02	-1.22
NumCompaniesWorked	21	1470	2.69	2.50	2.0	2.36	1.48	0	9	9	1.02	0.00
Over18*	22	1470	1.00	0.00	1.0	1.00	0.00	1	1	0	NaN	NaN
OverTime*	23	1470	1.28	0.45	1.0	1.23	0.00	1	2	1	0.96	-1.07
PercentSalaryHike	24	1470	15.21	3.66	14.0	14.80	2.97	11	25	14	0.82	-0.31
PerformanceRating	25	1470	3.15	0.36	3.0	3.07	0.00	3	4	1	1.92	1.68
RelationshipSatisfaction	26	1470	2.71	1.08	3.0	2.77	1.48	1	4	3	-0.30	-1.19
StandardHours	27	1470	80.00	0.00	80.0	80.00	0.00	80	80	0	NaN	NaN
StockOptionLevel	28	1470	0.79	0.85	1.0	0.67	1.48	0	3	3	0.97	0.35
TotalWorkingYears	29	1470	11.28	7.78	10.0	10.37	5.93	0	40	40	1.11	0.91
TrainingTimesLastYear	30	1470	2.80	1.29	3.0	2.72	1.48	0	6	6	0.55	0.48
WorkLifeBalance	31	1470	2.76	0.71	3.0	2.77	0.00	1	4	3	-0.55	0.41
YearsAtCompany	32	1470	7.01	6.13	5.0	5.99	4.45	0	40	40	1.76	3.91
YearsInCurrentRole	33	1470	4.23	3.62	3.0	3.85	4.45	0	18	18	0.92	0.47
YearsSinceLastPromotion	34	1470	2.19	3.22	1.0	1.48	1.48	0	15	15	1.98	3.59
YearsWithCurrManager	35	1470	4.12	3.57	3.0	3.77	4.45	0	17	17	0.83	0.16

	se
Age	0.24
Attrition*	0.01
BusinessTravel*	0.02
DailyRate	10.52
Department*	0.01
DistanceFromHome	0.21

```
> summary(df)
      Age          Attrition          BusinessTravel        DailyRate        Department         DistanceFromHome
 Min.   :18.00   Length:1470        Length:1470        Min.   : 102.0   Length:1470        Min.   : 1.000
 1st Qu.:30.00   Class :character   Class :character   1st Qu.: 465.0   Class :character   1st Qu.: 2.000
 Median :36.00   Mode  :character   Mode  :character   Median : 802.0   Mode  :character   Median : 7.000
 Mean   :36.92                                         Mean   : 802.5                      Mean   : 9.193
 3rd Qu.:43.00                                         3rd Qu.:1157.0                      3rd Qu.:14.000
 Max.   :60.00                                         Max.   :1499.0                      Max.   :29.000
   Education     EducationField     EmployeeCount EmployeeNumber   EnvironmentSatisfaction    Gender
 Min.   :1.000   Length:1470        Min.   :1     Min.   :   1.0   Min.   :1.000           Length:1470
 1st Qu.:2.000   Class :character   1st Qu.:1     1st Qu.: 491.2   1st Qu.:2.000           Class :character
 Median :3.000   Mode  :character   Median :1     Median :1020.5   Median :3.000           Mode  :character
 Mean   :2.913                      Mean   :1     Mean   :1024.9   Mean   :2.722
 3rd Qu.:4.000                      3rd Qu.:1     3rd Qu.:1555.8   3rd Qu.:4.000
 Max.   :5.000                      Max.   :1     Max.   :2068.0   Max.   :4.000
   HourlyRate     JobInvolvement    JobLevel       JobRole          JobSatisfaction MaritalStatus
 Min.   : 30.00   Min.   :1.00   Min.   :1.000   Length:1470        Min.   :1.000   Length:1470
 1st Qu.: 48.00   1st Qu.:2.00   1st Qu.:1.000   Class :character   1st Qu.:2.000   Class :character
 Median : 66.00   Median :3.00   Median :2.000   Mode  :character   Median :3.000   Mode  :character
 Mean   : 65.89   Mean   :2.73   Mean   :2.064                      Mean   :2.729
 3rd Qu.: 83.75   3rd Qu.:3.00   3rd Qu.:3.000                      3rd Qu.:4.000
 Max.   :100.00   Max.   :4.00   Max.   :5.000                      Max.   :4.000
 MonthlyIncome    MonthlyRate    NumCompaniesWorked    Over18            OverTime         PercentSalaryHike
 Min.   : 1009   Min.   : 2094   Min.   :0.000      Length:1470        Length:1470        Min.   :11.00
 1st Qu.: 2911   1st Qu.: 8047   1st Qu.:1.000      Class :character   Class :character   1st Qu.:12.00
 Median : 4919   Median :14236   Median :2.000      Mode  :character   Mode  :character   Median :14.00
 Mean   : 6503   Mean   :14313   Mean   :2.693                                            Mean   :15.21
 3rd Qu.: 8379   3rd Qu.:20462   3rd Qu.:4.000                                            3rd Qu.:18.00
 Max.   :19999   Max.   :26999   Max.   :9.000                                            Max.   :25.00
 PerformanceRating RelationshipSatisfaction StandardHours StockOptionLevel TotalWorkingYears
 Min.   :3.000     Min.   :1.000            Min.   :80    Min.   :0.0000   Min.   : 0.00
 1st Qu.:3.000     1st Qu.:2.000            1st Qu.:80    1st Qu.:0.0000   1st Qu.: 6.00
 Median :3.000     Median :3.000            Median :80    Median :1.0000   Median :10.00
 Mean   :3.154     Mean   :2.712            Mean   :80    Mean   :0.7939   Mean   :11.28
 3rd Qu.:3.000     3rd Qu.:4.000            3rd Qu.:80    3rd Qu.:1.0000   3rd Qu.:15.00
 Max.   :4.000     Max.   :4.000            Max.   :80    Max.   :3.0000   Max.   :40.00
 TrainingTimesLastYear WorkLifeBalance YearsAtCompany   YearsInCurrentRole YearsSinceLastPromotion
 Min.   :0.000         Min.   :1.000   Min.   : 0.000   Min.   : 0.000     Min.   : 0.000
 1st Qu.:2.000         1st Qu.:2.000   1st Qu.: 3.000   1st Qu.: 2.000     1st Qu.: 0.000
 Median :3.000         Median :3.000   Median : 5.000   Median : 3.000     Median : 1.000
 Mean   :2.799         Mean   :2.761   Mean   : 7.008   Mean   : 4.229     Mean   : 2.188
 3rd Qu.:3.000         3rd Qu.:3.000   3rd Qu.: 9.000   3rd Qu.: 7.000     3rd Qu.: 3.000
 Max.   :6.000         Max.   :4.000   Max.   :40.000   Max.   :18.000     Max.   :15.000
 YearsWithCurrManager
 Min.   : 0.000
```

- Data is split into numerical and categorical variables.
- Categorical variables are converted to factors.

3. Methodology

The project employs several AI/ML techniques suitable for binary classification tasks:

- **Logistic Regression:** A statistical method for analyzing a dataset in which there are one or more independent variables that determine an outcome.

- **Random Forests:** An ensemble learning method for classification that operates by constructing multiple decision trees during training.
- **Gradient Boosting:** A machine learning technique for regression and classification problems, which produces a prediction model in the form of an ensemble of weak prediction models.
- **CART (Classification and Regression Tree):** A decision tree methodology that provides a way to model decisions and their possible consequences.

Tools and Libraries:

- R programming language
- Libraries such as readr, dplyr, psych, ggplot2, caret, randomForest, pROC, gbm, rpart, rpart.plot, and glmnet.

4. Exploratory Analysis

4.1 Data Exploration

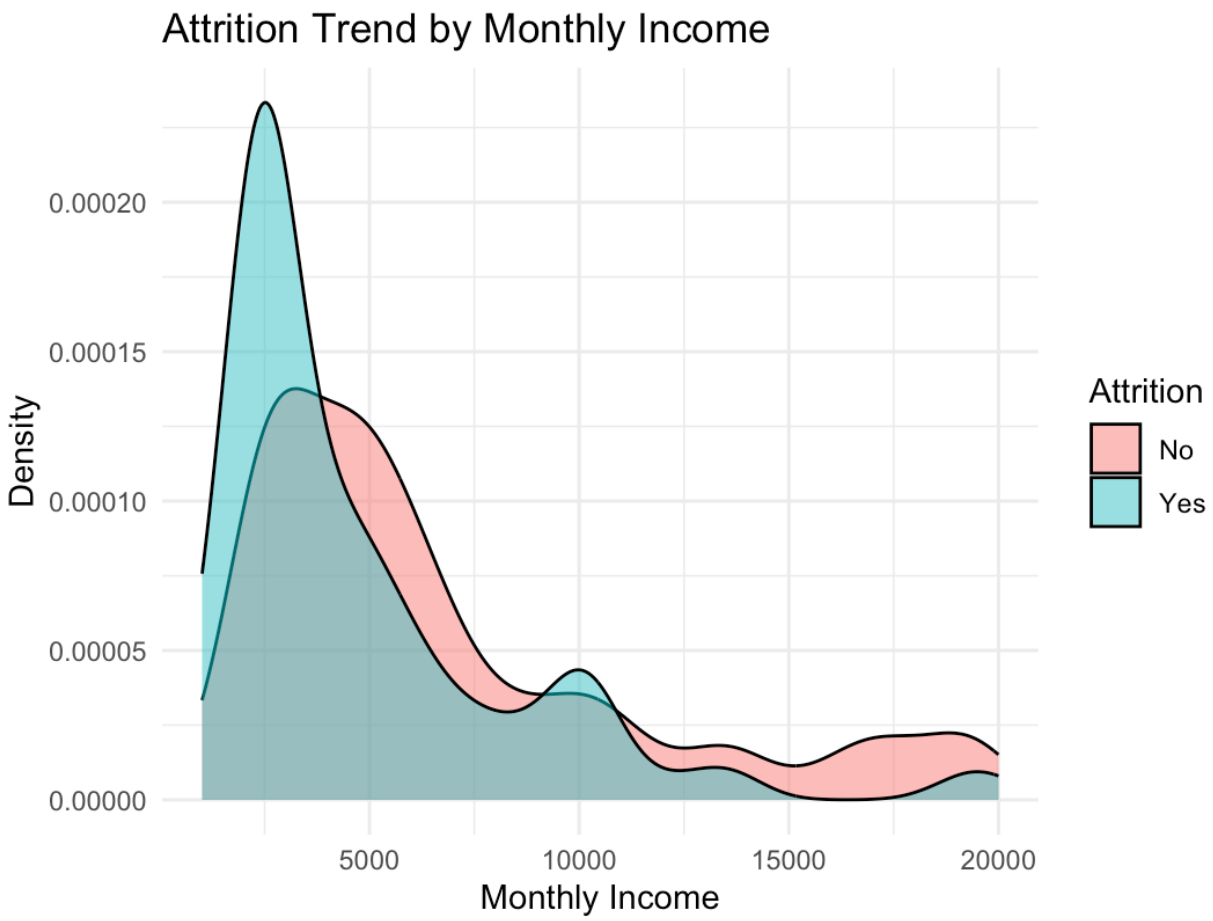
The dataset contains 1470 rows and 35 columns, with no missing values or duplicates. Initial exploration included checking dimensions, basic statistics, and summaries for all variables.

Correlation Matrix: Correlation matrix has been generated among numerical variables and top correlations are printed.

```
> print(top_correlations)
                 Var1                     Var2 Correlation
1            JobLevel            MonthlyIncome   0.9502999
2            JobLevel        TotalWorkingYears   0.7822078
3   PercentSalaryHike        PerformanceRating   0.7735500
4       MonthlyIncome        TotalWorkingYears   0.7728932
5      YearsAtCompany     YearsWithCurrManager   0.7692124
6      YearsAtCompany       YearsInCurrentRole   0.7587537
7  YearsInCurrentRole     YearsWithCurrManager   0.7143648
8                 Age        TotalWorkingYears   0.6803805
9   TotalWorkingYears           YearsAtCompany   0.6281332
10     YearsAtCompany YearsSinceLastPromotion   0.6184089
```

4.2 Key Visualizations

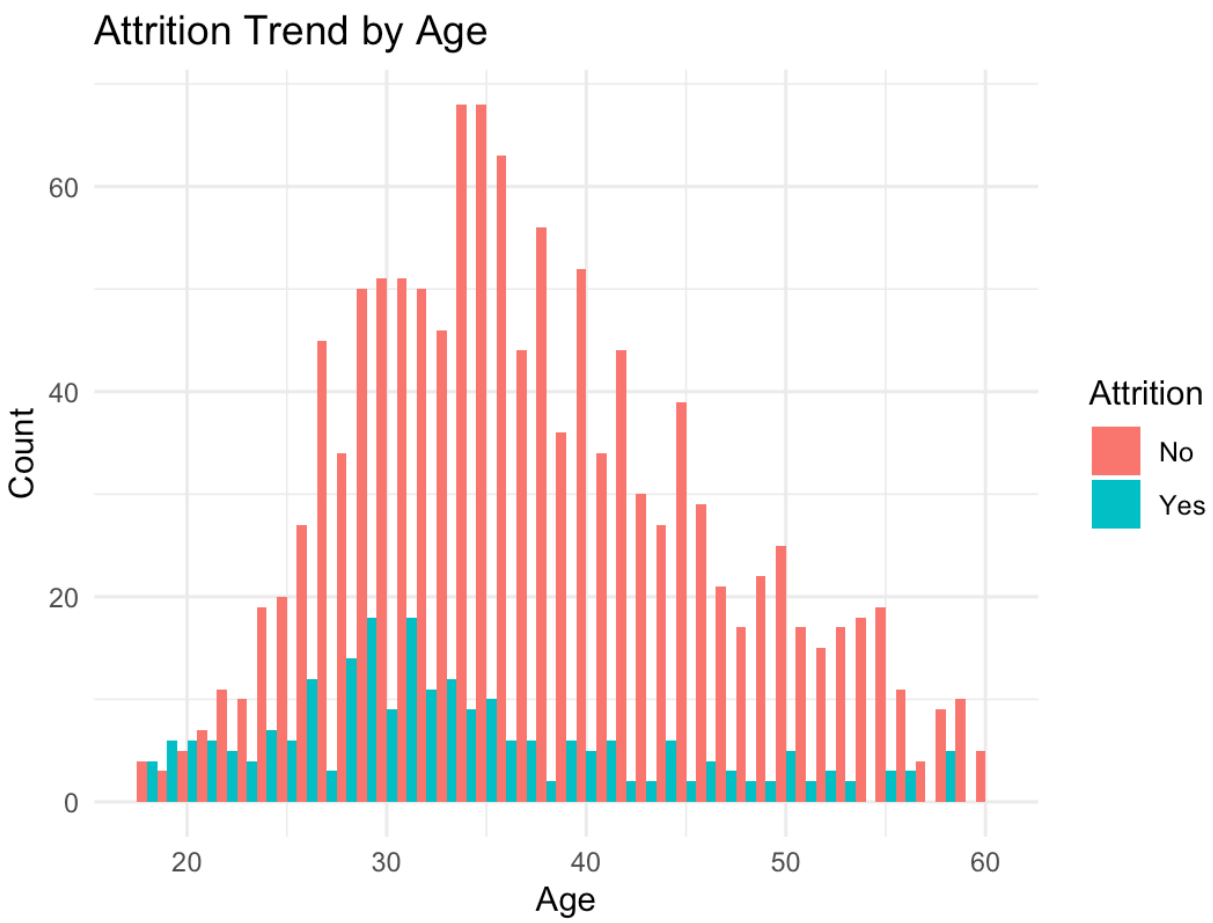
- **Attrition Trend by Monthly Income:**



  - Lower-income employees are more likely to leave the company. As income increases, the likelihood of attrition decreases.

  - **Insight:** Organizations should consider competitive compensation strategies to retain lower-income employees.

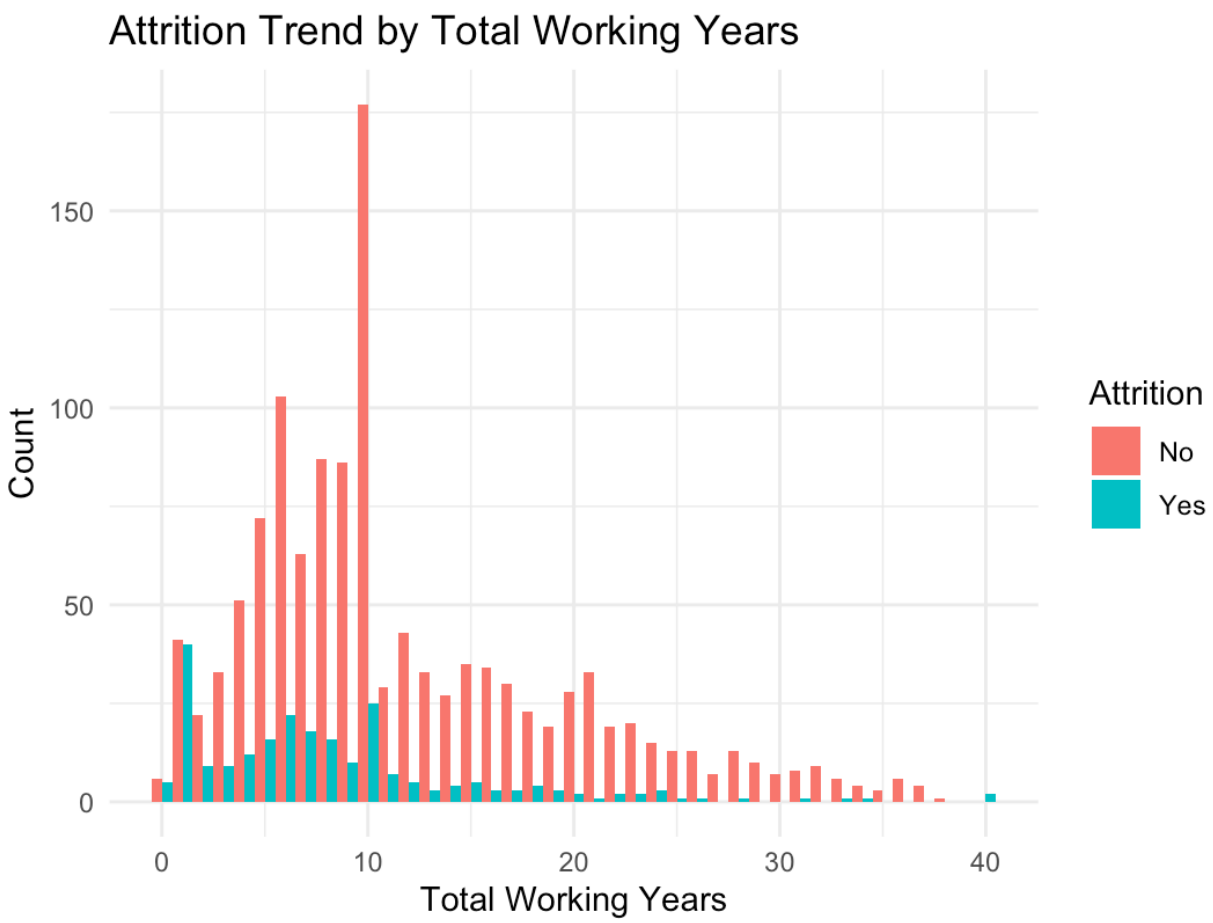
- **Attrition Trend by Age:**



  - Younger employees (early twenties to mid-thirties) have higher attrition rates. Middle-aged employees (late thirties to late forties) are more stable, while attrition slightly increases for employees over 50.

  - **Insight:** Tailored engagement and career development opportunities for younger employees could help in retaining them.

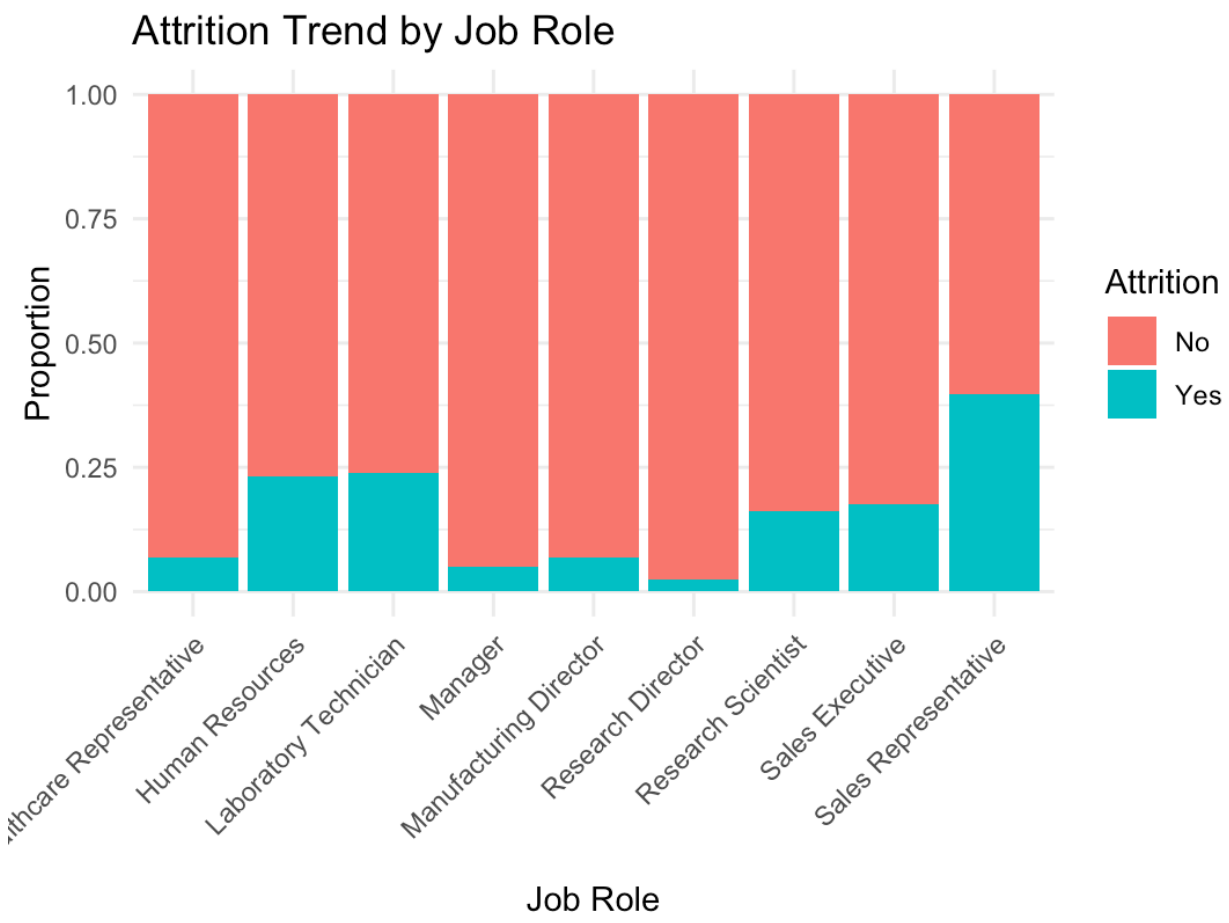
- **Attrition Trend by Total Working Years:**



  - Employees with fewer than 10 working years have higher attrition rates. Attrition decreases for those with 10-20 years of experience and remains low for employees with more than 20 years.

  - **Insight:** Providing clear career progression paths and opportunities for skill development can help reduce attrition among early-career employees.

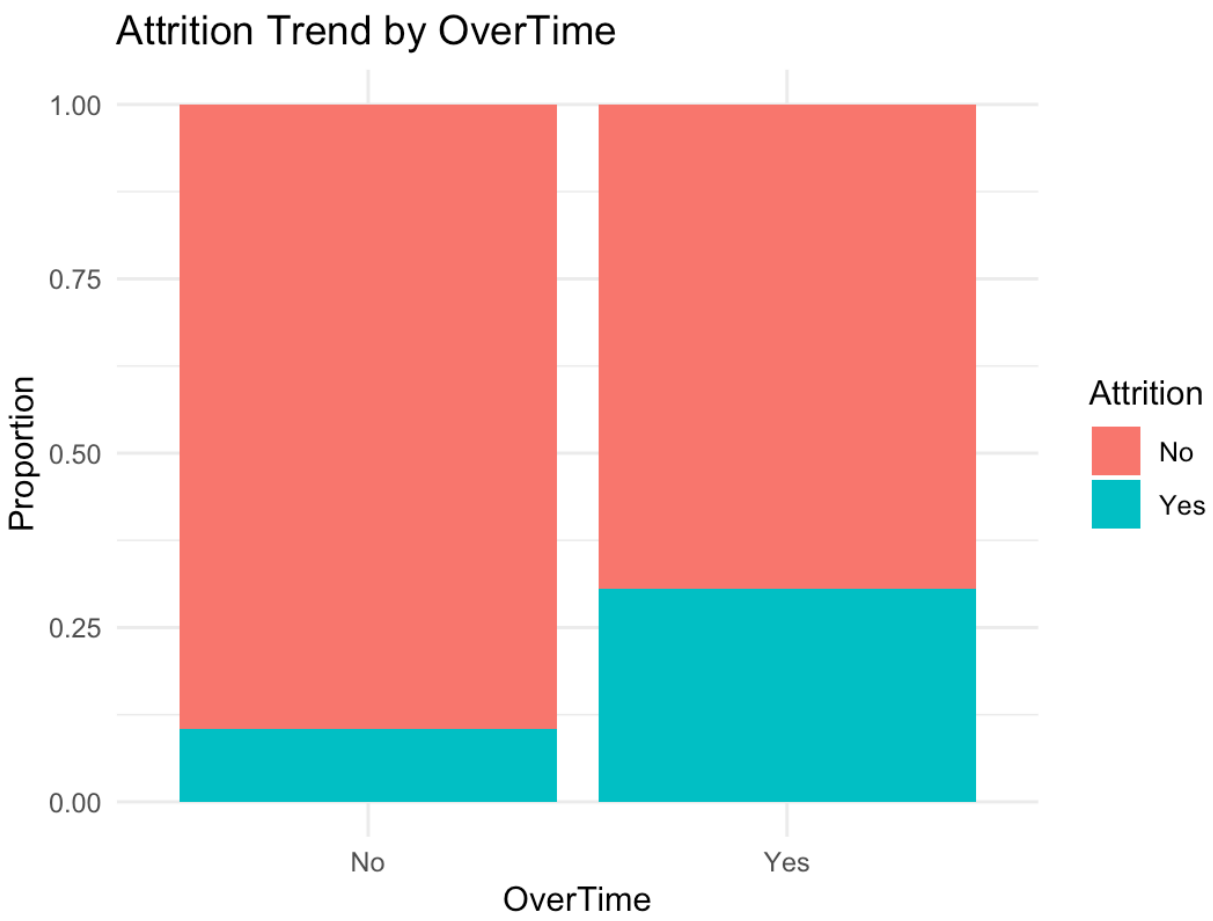
- **Attrition Trend by Job Role:**



  - Sales positions (Sales Representatives and Sales Executives) have the highest attrition rates. Healthcare Representatives, HR, and Managers have lower attrition rates.

  - **Insight:** Addressing specific challenges and providing support in high-turnover roles like sales can improve retention.

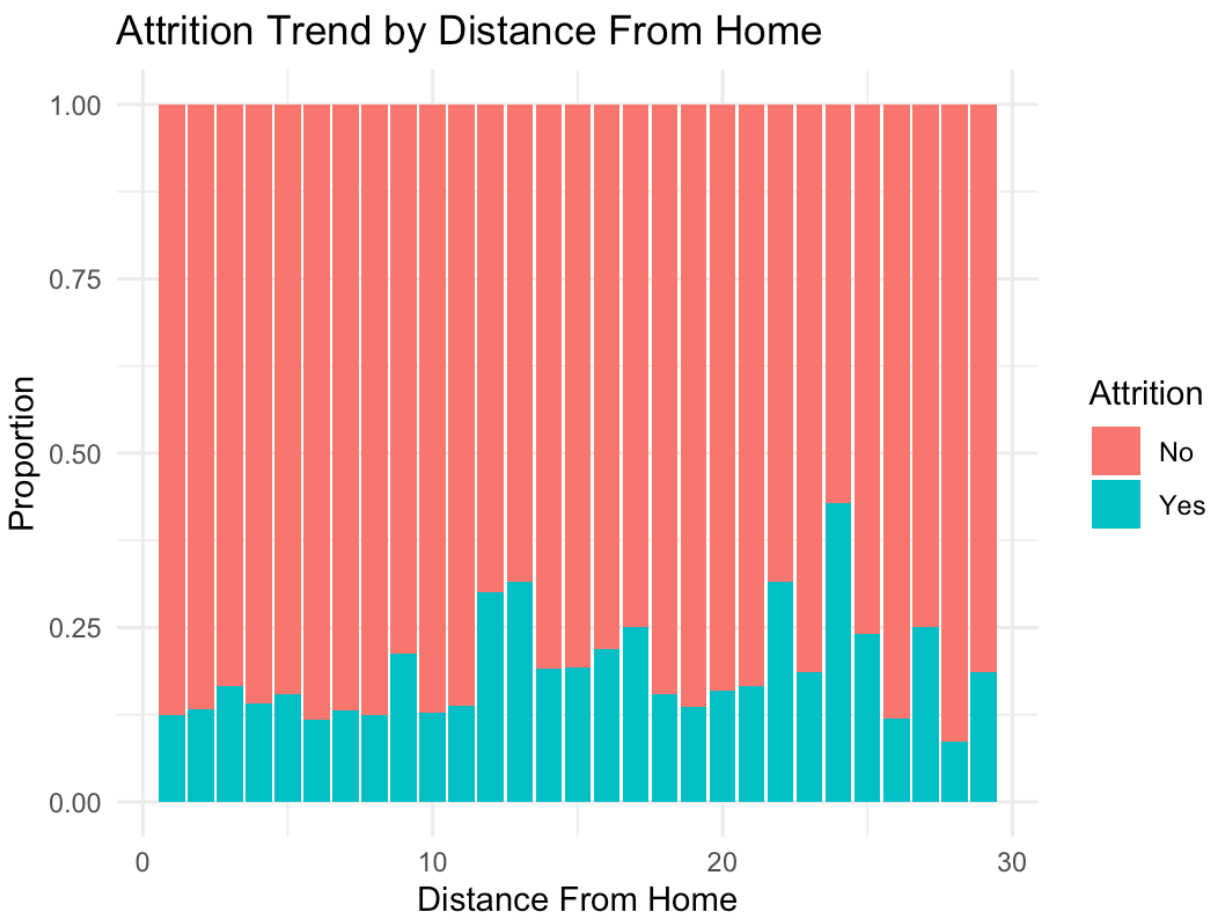
- **Attrition Trend by OverTime:**



  - Employees working overtime have significantly higher attrition rates compared to those who do not.

  - **Insight:** Managing overtime and ensuring work-life balance is crucial for employee retention.

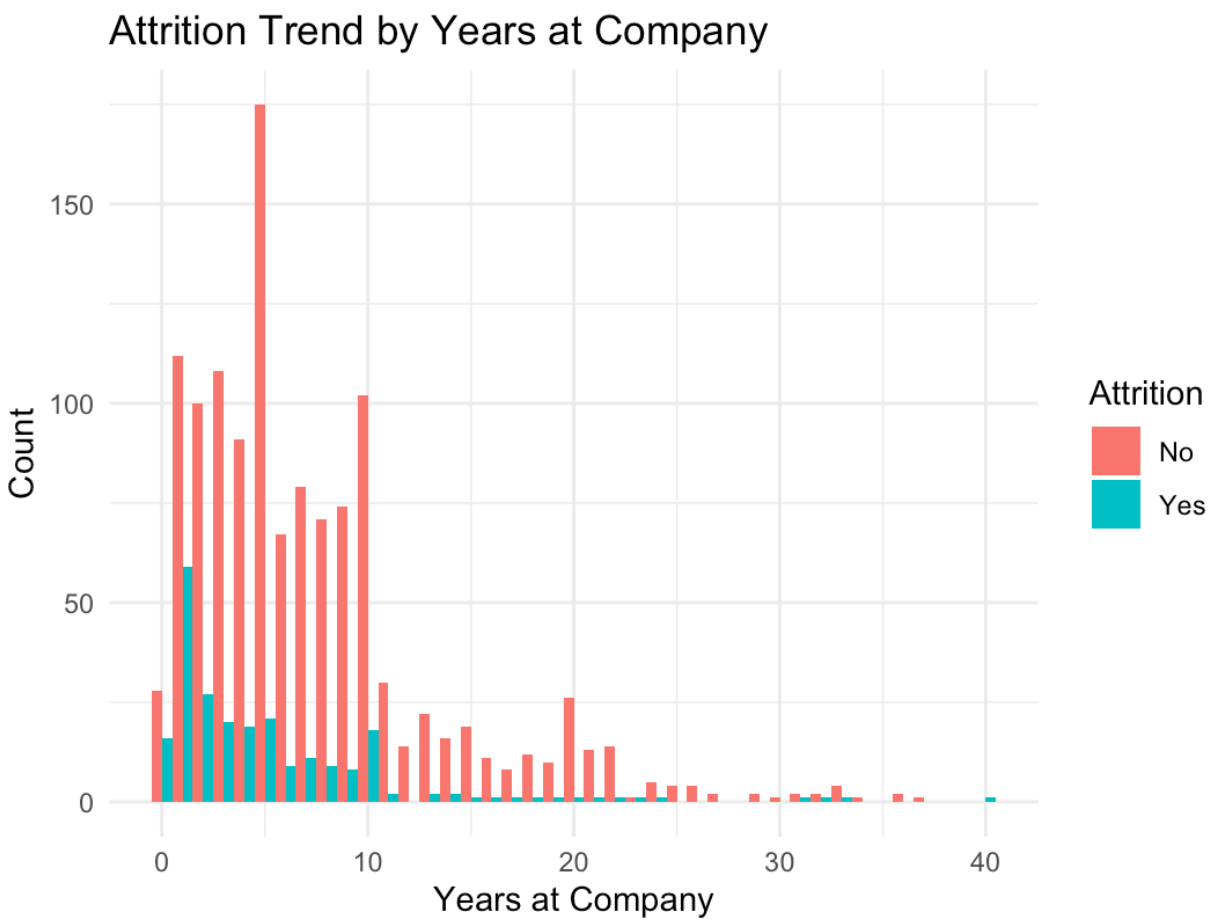
- **Attrition Trend by Distance from Home:**



  - Employees living closer to work have lower attrition rates. As commute distance increases, so does the likelihood of attrition.

  - **Insight:** Offering flexible working arrangements or transportation benefits could help reduce attrition for employees with long commutes.

- **Attrition Trend by Years at Company:**



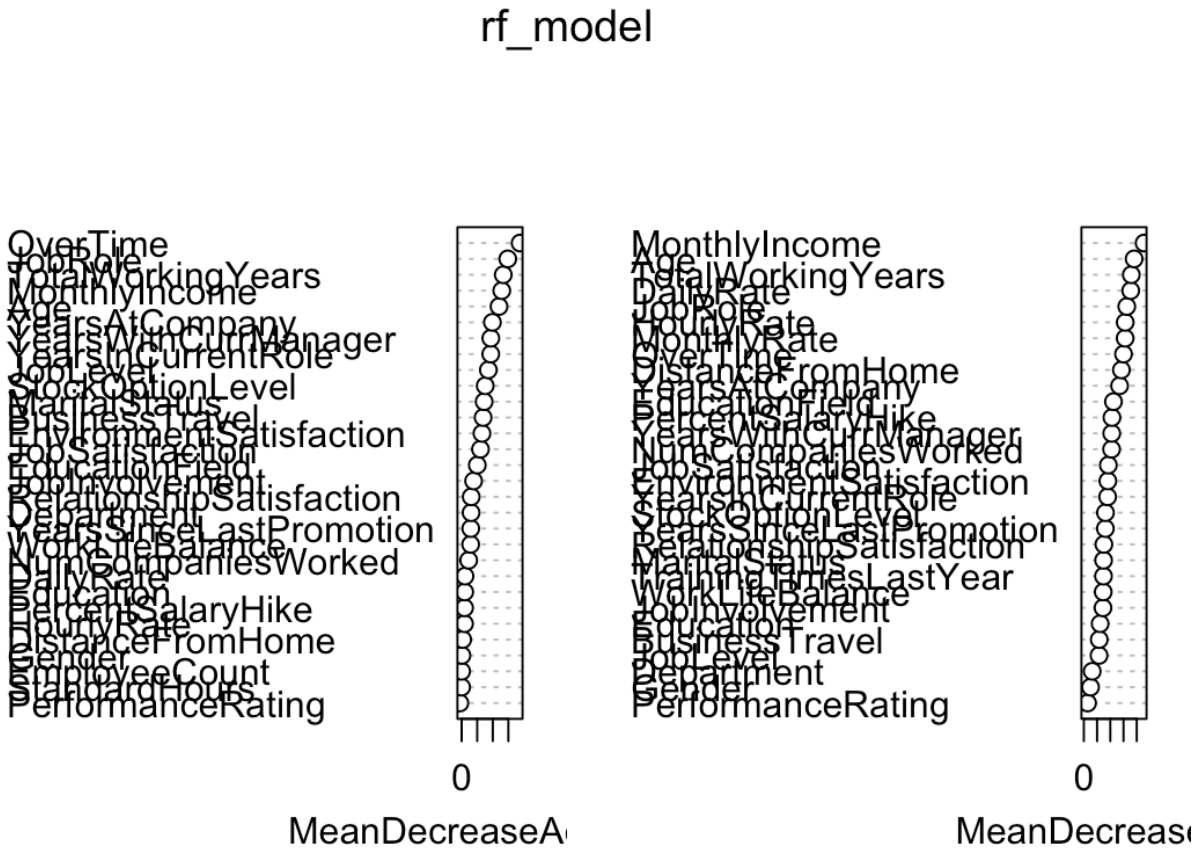
  - Employees in their first few years at the company show higher attrition rates. Attrition decreases significantly for employees with 3-10 years at the company and remains low for those with longer tenure.

  - **Insight:** Focusing on the onboarding experience and early career support can help in retaining new employees.

5. Model Building and Evaluation

5.1 Random Forest

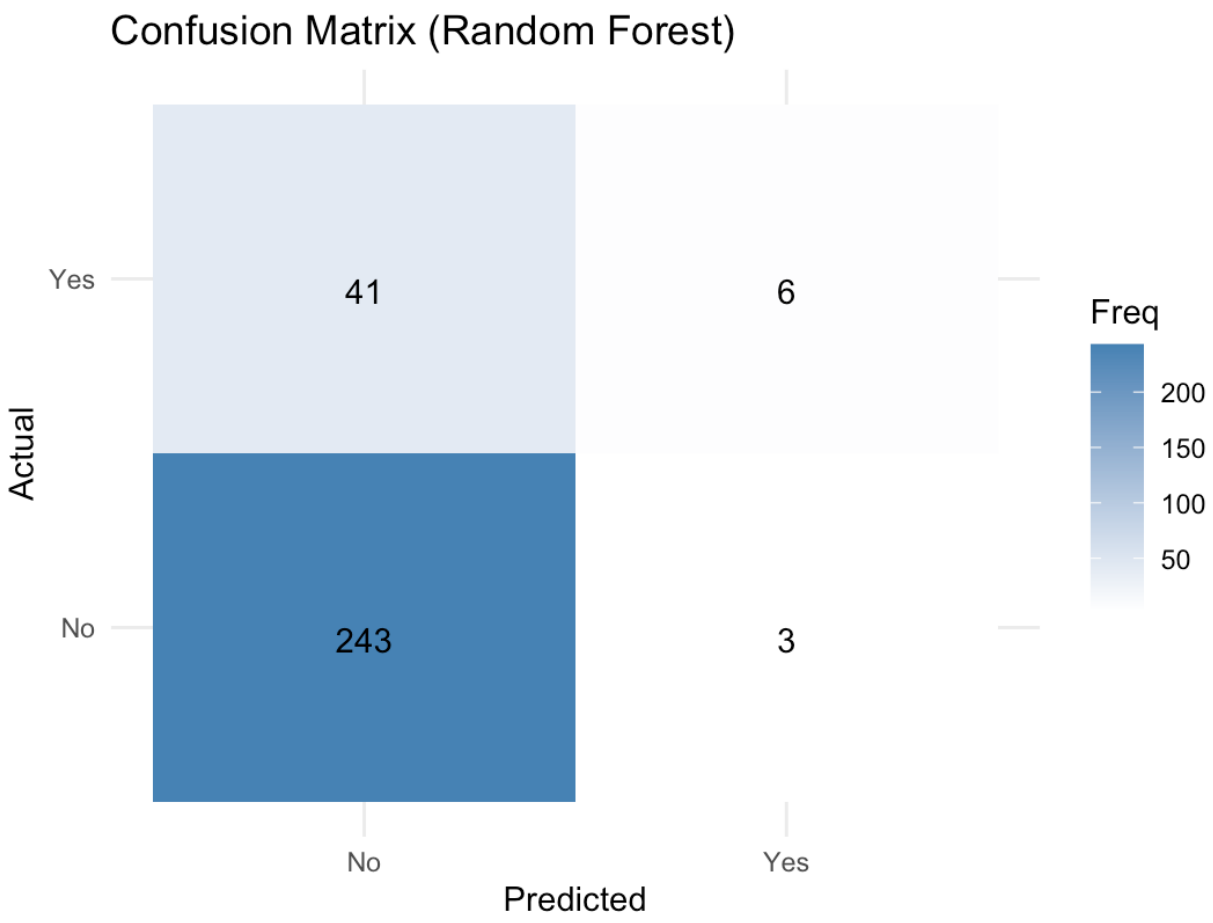
- **Feature Importance:**
  - Key features include OverTime, Job Role, Total Working Years, and Monthly Income as per MeanDecreaseAccuracy.



```
> # Variable importance
> rf_importance <- randomForest::importance(rf_model)
> print(rf_importance)
                                 No        Yes MeanDecreaseAccuracy MeanDecreaseGini
Age                       8.71102613  8.2343213           12.0062902        19.008913
BusinessTravel            4.62186479  5.0704837            6.8981813         5.815923
DailyRate                 1.69761471 -0.8215713            1.0766329        17.720363
Department                1.57751850  2.9421567            3.0348222         3.163589
DistanceFromHome         -0.64531173  1.7391588            0.3384187        14.205911
Education                 0.83957577  0.5817767            0.9838500         6.156720
EducationField            4.41062357  2.4035199            5.0290757        11.201393
EmployeeCount             0.00000000  0.0000000            0.0000000         0.000000
EnvironmentSatisfaction   4.86458259  4.4696112            6.5289436         9.079148
Gender                    0.21283914 -0.1644333            0.1004179         2.525225
HourlyRate                0.62378391  0.8007196            0.9225672        15.636636
JobInvolvement            3.47493143  1.4439412            3.8374304         7.236123
JobLevel                  7.56021063  2.3054770            8.5745423         5.739924
JobRole                  12.82926409  7.5511457           14.8112410        16.234905
JobSatisfaction           3.93298736  5.7856531            6.0469464         9.260966
MaritalStatus             4.35923750  7.8047417            7.2339542         7.461611
MonthlyIncome            10.61176443  4.8800476           12.8830115        22.791886
MonthlyRate              -3.00682016 -1.4030904           -3.3410505        15.610463
NumCompaniesWorked        2.13596939  0.6831758            2.2626826        10.478466
OverTime                 13.78540200 17.1178135           18.7808460        15.002730
PercentSalaryHike         1.26892762 -0.4268227            0.9381032        10.625812
PerformanceRating        -0.03991621 -1.1032568           -0.5755359         1.422299
RelationshipSatisfaction  3.25021343  0.6733714            3.0747910         7.579981
StandardHours             0.00000000  0.0000000            0.0000000         0.000000
StockOptionLevel          5.21359593  6.7231919            7.5679753         8.179726
TotalWorkingYears        11.77156130  3.5072999           13.3725937        17.939803
TrainingTimesLastYear    -0.78006597 -1.4927942           -1.3367150         7.418085
WorkLifeBalance           1.75391608  2.7178183            2.8999423         7.326663
YearsAtCompany            7.65098009  5.6280553            9.8850845        13.508194
YearsInCurrentRole        7.69153331  4.1593304            9.3514168         8.999092
YearsSinceLastPromotion   3.26505970 -0.2458022            2.9265285         8.140730
YearsWithCurrManager      7.13850299  5.8964754            9.3746134        10.544386
```

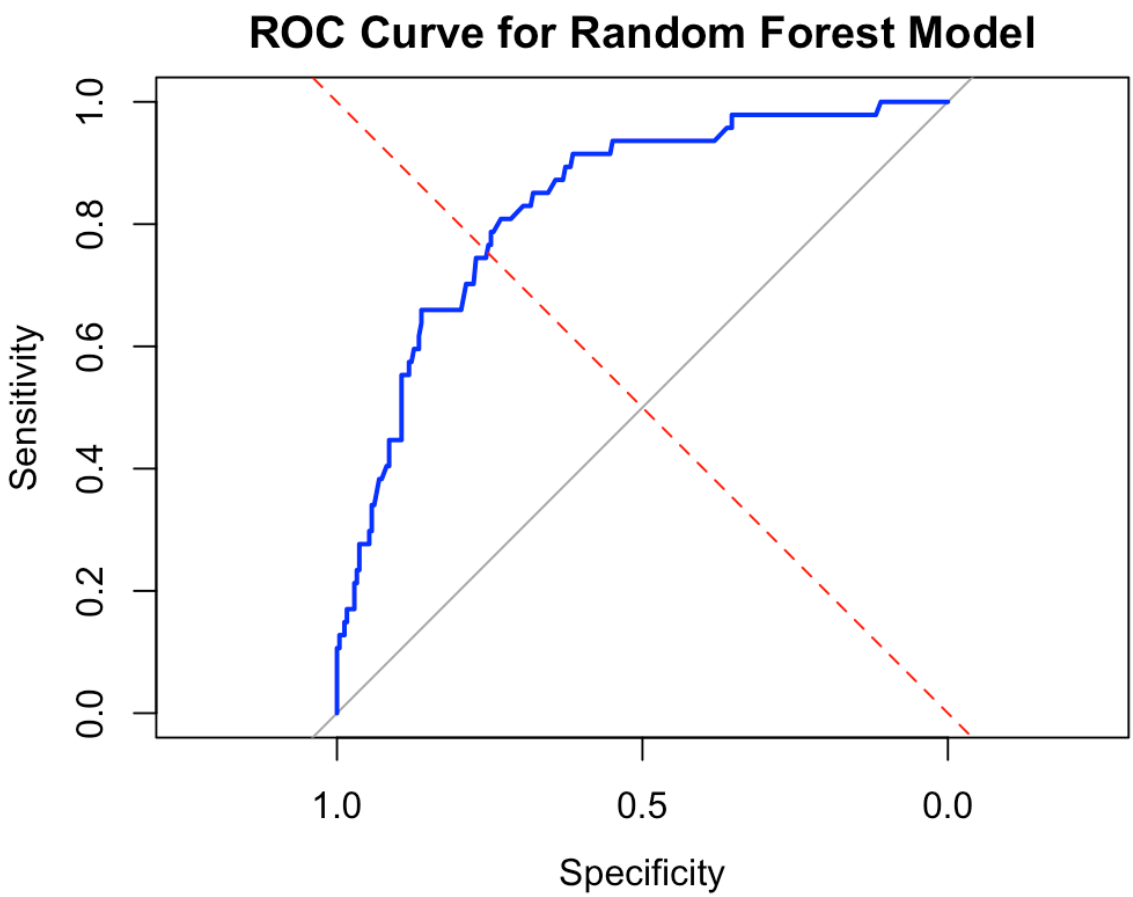
```
> # Feature selection based on importance
> # Extract top 10 important features
> important_features <- rownames(rf_importance)[order(rf_importance[, "MeanDecreaseGini"], decreasing = TRUE)][1:10]
> print(important_features)
 [1] "MonthlyIncome"     "Age"               "TotalWorkingYears" "DailyRate"         "JobRole"
 [6] "HourlyRate"        "MonthlyRate"       "OverTime"          "DistanceFromHome"  "YearsAtCompany"
```

- **Confusion Matrix:**



  - The model correctly predicts a high number of true negatives but has some false positives and false negatives.

- **ROC Curve:**

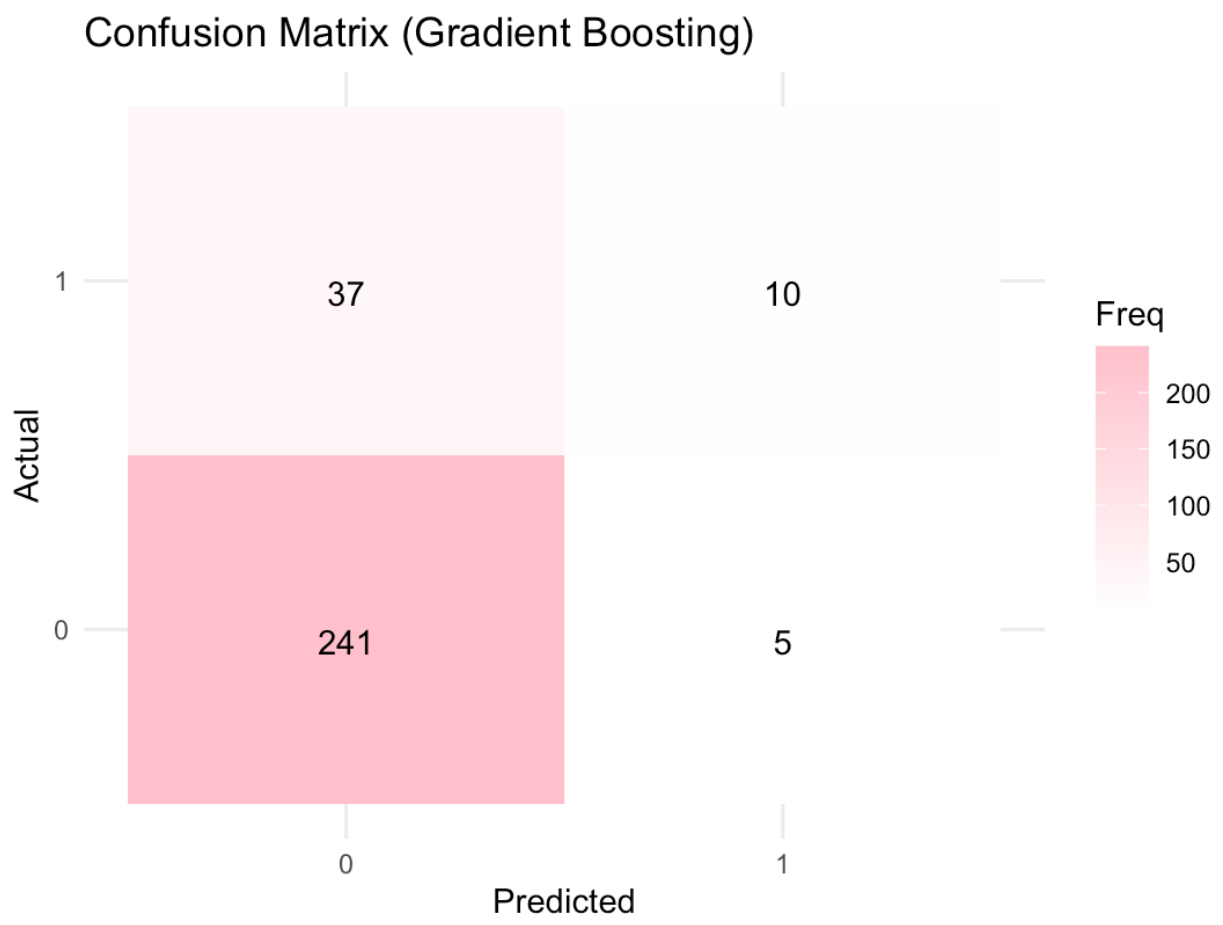


  - The ROC curve shows good performance with a high AUC of approximately 0.83.

  - **Insight:** Random Forest is effective with a good true negative rate but requires improvement in reducing false positives and false negatives.

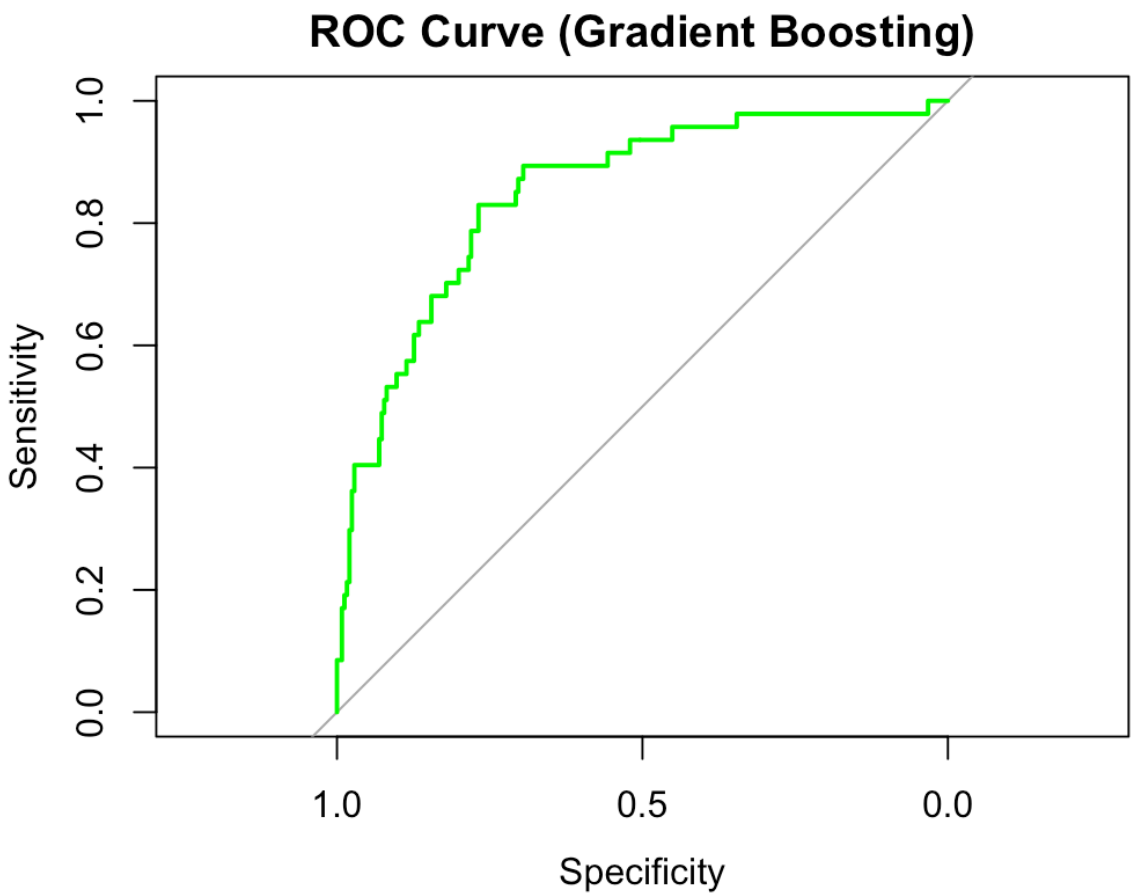
5.2 Gradient Boosting

- **Confusion Matrix:**



  - The model has a balanced performance with a moderate number of true positives and low false predictions.

- **ROC Curve:**

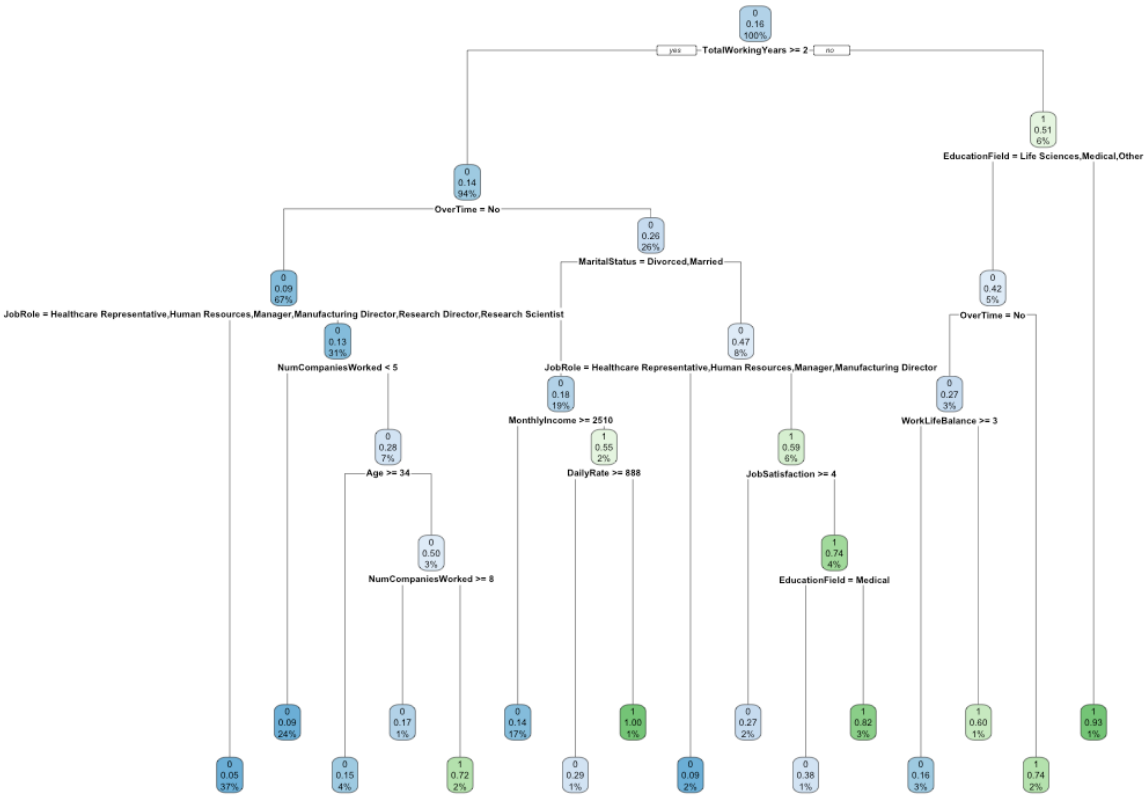


  - The ROC curve indicates good performance with a high AUC of approximately 0.85.

  - **Insight:** Gradient Boosting shows high predictive accuracy with a well-balanced performance, making it reliable for predicting employee attrition.

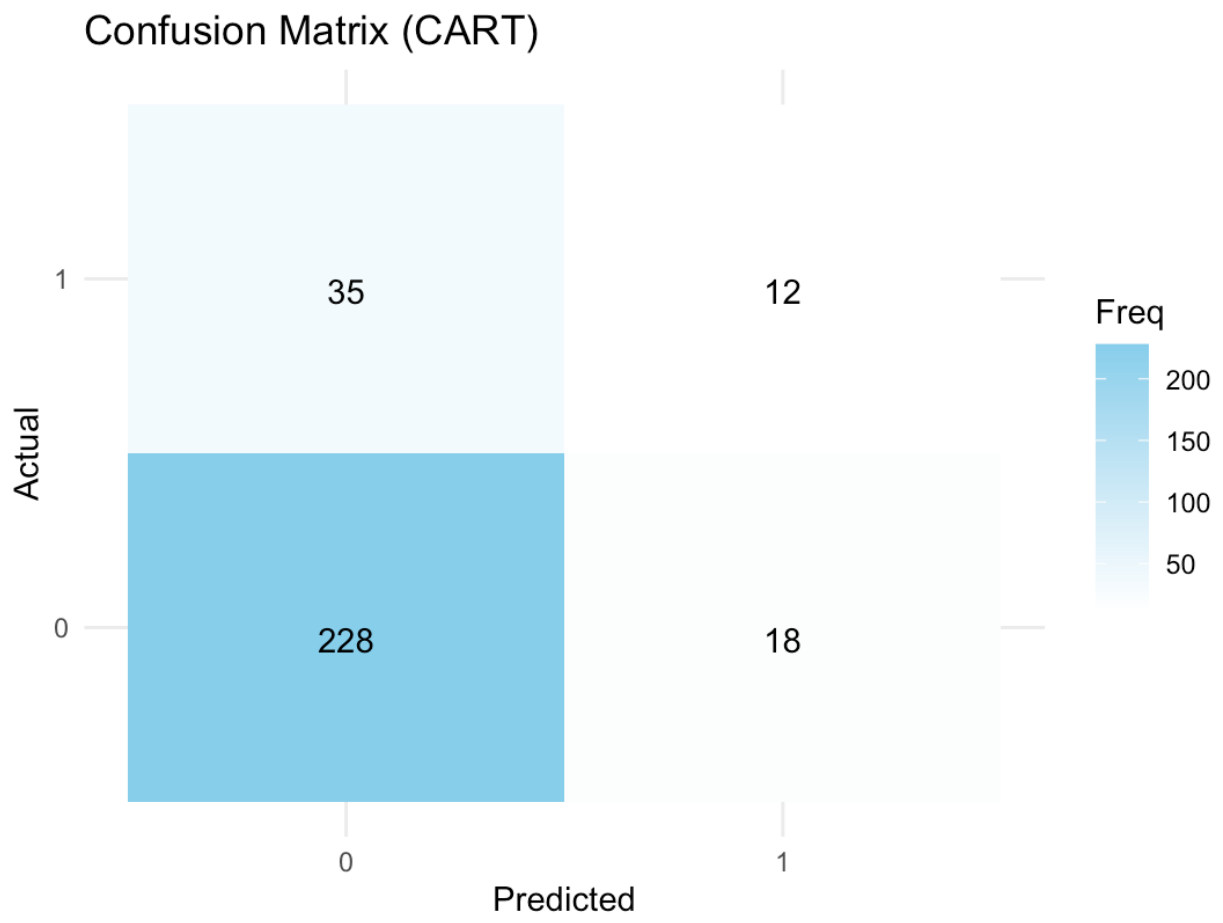
5.3 CART

- **Decision Tree Visualization:**



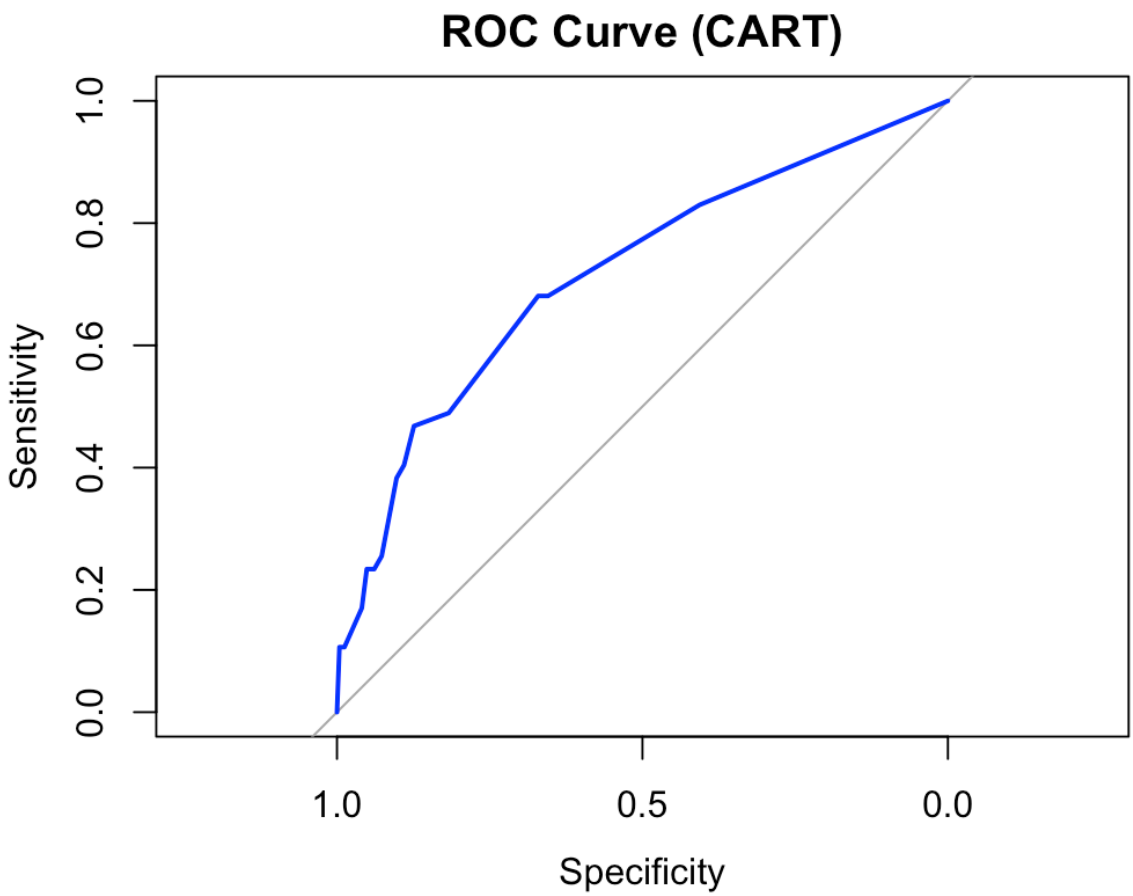
- The decision tree provides insights on factors such as OverTime, Total Working Years, Job Role, and Work-Life Balance.

- **Confusion Matrix:**



  - The model performs moderately well with a good number of true negatives but some false positives and false negatives.

- **ROC Curve:**

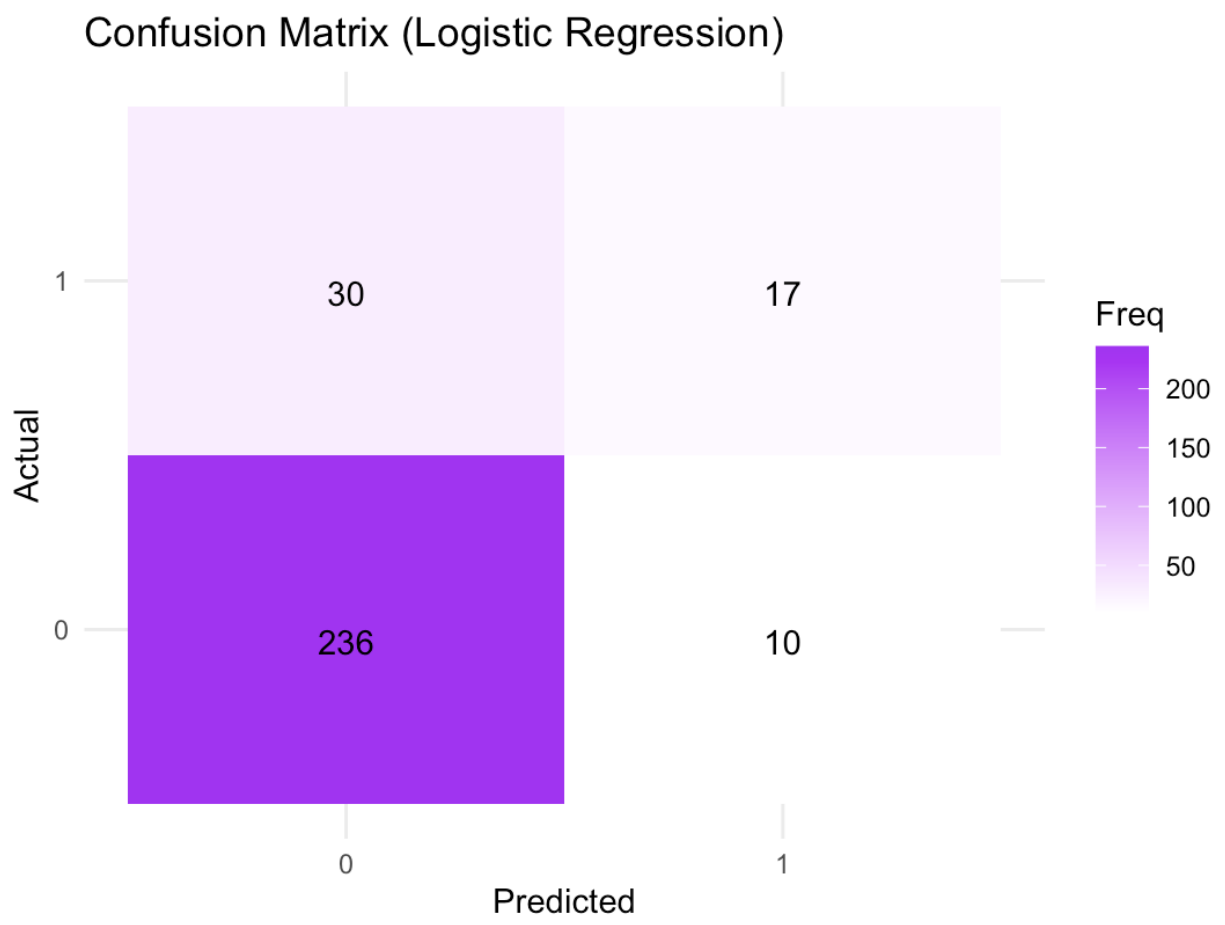


  - The ROC curve shows moderate performance with an AUC of approximately 0.71.

  - **Insight:** CART provides clear, interpretable decision rules but may need further tuning for better accuracy.

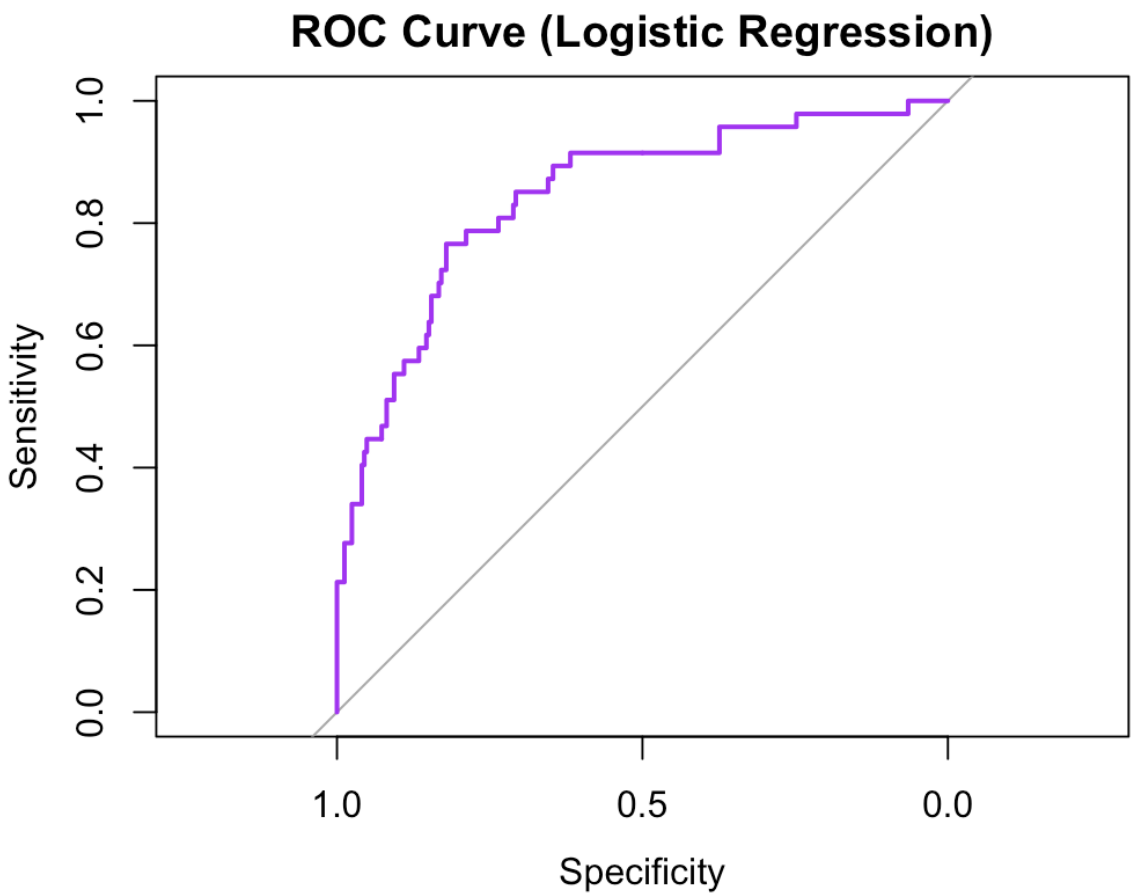
5.4 Logistic Regression

- **Confusion Matrix:**



  - The model performs well with a balanced number of true positives and low false predictions.

- **ROC Curve:**



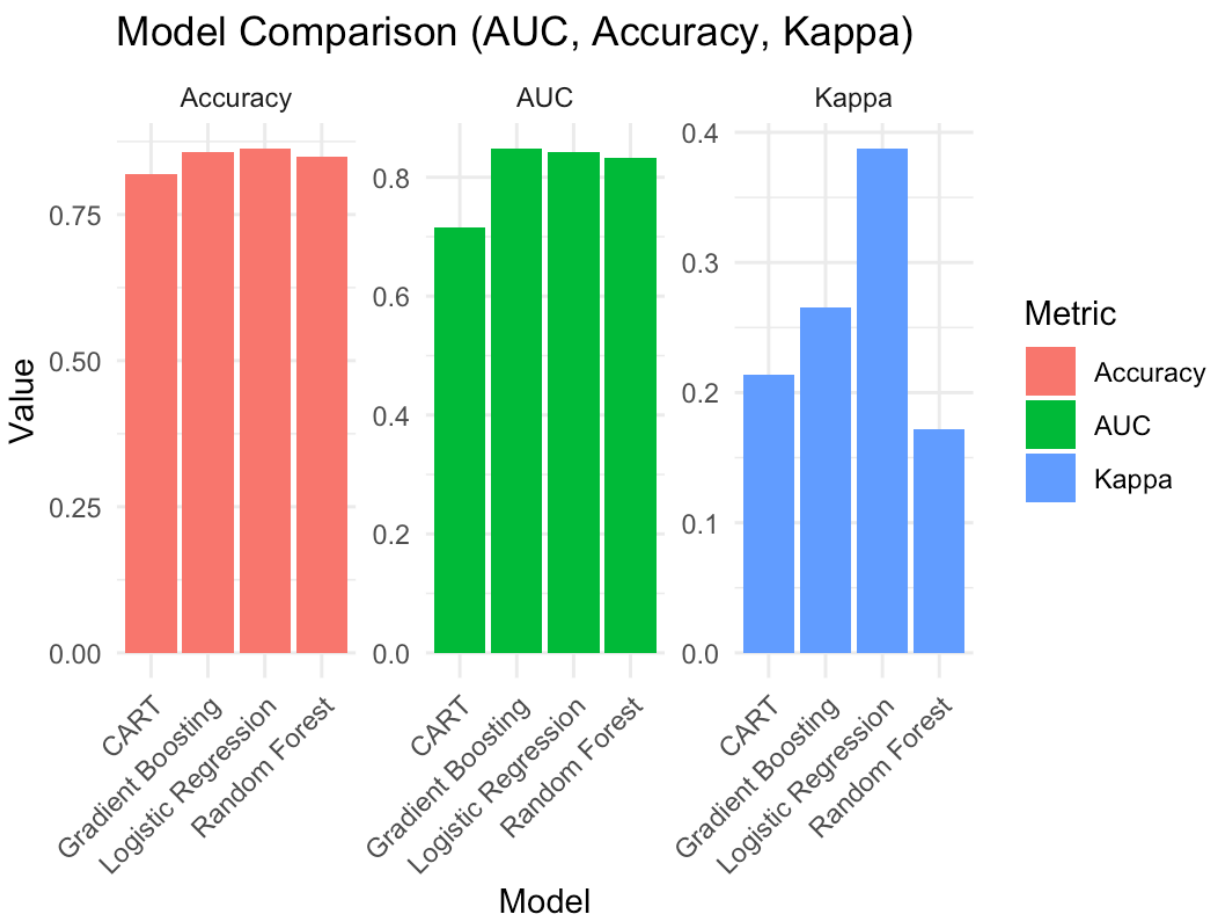
  - The ROC curve shows good performance with a high AUC of approximately 0.84.

  - **Insight:** Logistic Regression is a strong model with high accuracy and balanced performance, making it suitable for predicting employee attrition.

6. Model Comparison

6.1 AUC, Accuracy, and Kappa

```
> # Print AUC, Accuracy, and Kappa values
> print("Model Comparison - AUC Values:")
[1] "Model Comparison - AUC Values:"
> print(auc_values)
      Random Forest   Gradient Boosting                CART Logistic Regression
          0.8324252           0.8488151           0.7153174           0.8431932
> print("Model Comparison - Accuracy Values:")
[1] "Model Comparison - Accuracy Values:"
> print(accuracy_values)
      Random Forest.Accuracy   Gradient Boosting.Accuracy                CART.Accuracy
                   0.8498294                    0.8566553                    0.8191126
Logistic Regression.Accuracy
                   0.8634812
> print("Model Comparison - Kappa Values:")
[1] "Model Comparison - Kappa Values:"
> print(kappa_values)
      Random Forest.Kappa   Gradient Boosting.Kappa                CART.Kappa Logistic Regression.Kappa
                0.1715718                 0.2655765                 0.2133631                 0.3877977
>
```

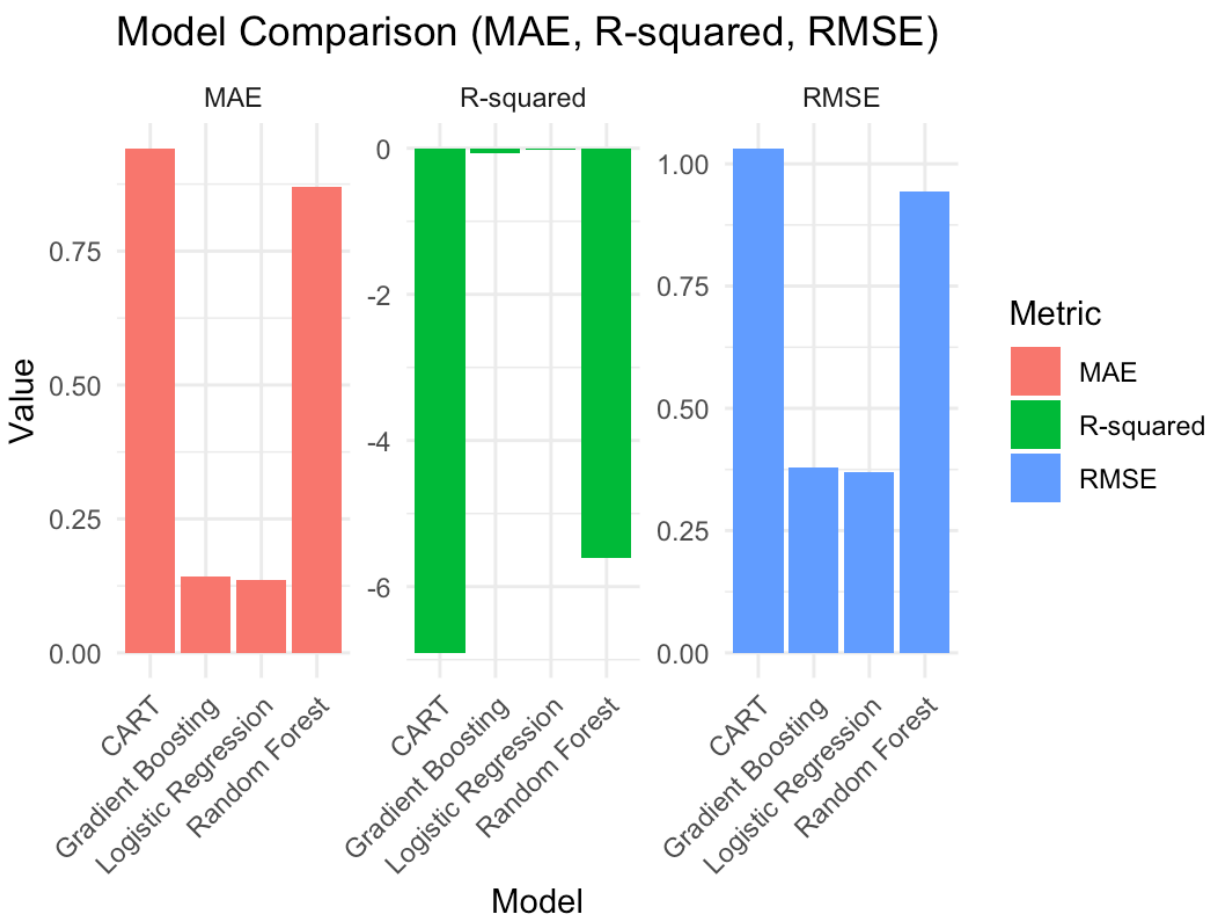


- **AUC (Area Under the Curve):**
  - Logistic Regression and Gradient Boosting show the highest AUC, indicating their effectiveness in predicting attrition.
- **Accuracy:**
  - All models have almost similar accuracy, above 0.80, indicating their overall reliability with Logistic Regression having highest as 0.86.
- **Kappa:**
  - Logistic Regression has the highest Kappa value (0.38), suggesting it performs significantly better than random guessing.

- **Insight:** Logistic Regression and Gradient Boosting are the most effective models based on AUC and Kappa, with Logistic Regression slightly leading in overall performance.

6.2 MAE, R-squared, and RMSE

```
> # Print MAE, RMSE, and R-squared values
> print("Model Comparison - MAE Values:")
[1] "Model Comparison - MAE Values:"
> print(mae_values)
      Random Forest.MAE   Gradient Boosting.MAE                CART.MAE Logistic Regression.MAE
              0.8703072               0.1433447               0.9419795               0.1365188
> print("Model Comparison - RMSE Values:")
[1] "Model Comparison - RMSE Values:"
> print(rmse_values)
      Random Forest.RMSE   Gradient Boosting.RMSE                CART.RMSE Logistic Regression.RMSE
               0.9438141                0.3786089                1.0319140                0.3694845
> print("Model Comparison - R-squared Values:")
[1] "Model Comparison - R-squared Values:"
> print(r2_values)
      Random Forest.R_squared   Gradient Boosting.R_squared                CART.R_squared
                  -5.61416710                  -0.06434873                   -6.90659056
Logistic Regression.R_squared
                  -0.01366546
>
```



- **MAE (Mean Absolute Error):**
  - MAE measures the average magnitude of errors in predictions, without considering their direction. Lower MAE values indicate better model performance.
  - Gradient Boosting and Logistic Regression have the lowest MAE values, indicating they make fewer prediction errors on average.
  - CART and Random Forest have higher MAE values, suggesting they are less accurate.
- **R-squared:**
  - R-squared indicates the proportion of the variance in the dependent variable that is predictable from the independent variables. Higher R-squared values indicate better model performance.
  - All models have negative R-squared values, which typically suggests poor model performance. However, this may be due to the specific characteristics of the dataset or the nature of the prediction task.
  - Gradient Boosting and Logistic Regression have the least negative R-squared values, indicating they explain the variance better than CART and Random Forest.

- **RMSE (Root Mean Square Error):**
    - RMSE measures the square root of the average of squared differences between prediction and actual observation. Lower RMSE values indicate better model performance.
    - Gradient Boosting and Logistic Regression have the lowest RMSE values, indicating they make smaller errors in predictions.
    - CART and Random Forest have higher RMSE values, suggesting they are less accurate.

- **Insight:** Gradient Boosting and Logistic Regression consistently show lower MAE and RMSE values, indicating better accuracy and fewer prediction errors. They also have the least negative R-squared values, suggesting they explain more variance in the data.

7. Recommendations and Conclusion

Based on the analysis and model evaluations, the Logistic Regression and Gradient Boosting models are the most effective for predicting employee attrition. Both models show high accuracy, AUC, and have the lowest error rates.

Key Recommendations:

- **Manage Overtime:** Focus on managing overtime and providing better work-life balance to reduce attrition.
- **Address High-Turnover Roles:** Address specific challenges faced by employees in high-turnover roles such as sales.
- **Early Career Support:** Offer support and development opportunities for employees in their early years at the company to improve retention.
- **Commute Solutions:** Consider offering flexible working arrangements, remote work options, or transportation benefits to employees who live farther away.

This data-driven approach will help HR teams implement targeted strategies to retain valuable employees and enhance overall organizational performance.

8. References

- **Dataset:** https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset
- **Random Forest:** https://cran.rproject.org/web/packages/randomForest/randomForest.pdf
- **Gradient Boosting Machine (Generalized Boosted Regression Models):** https://cran.r-project.org/web/packages/gbm/gbm.pdf
- **CART/ rpart:** https://cran.r-project.org/web/packages/rpart/rpart.pdf
- **Logistic Regression:** https://cran.r-project.org/web/packages/HSAUR3/vignettes/Ch_logistic_regression_glm.pdf